SUPERVISED MACHINE LEARNING: CLASSIFICATION

Optimizing Exoplanet Discovery: Automated Classification of Kepler Objects of Interest

IBM Machine Learning Course Final Project - Coursera
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AIM OF THE PROJECT

Classification of KOI into Exoplanets and Non Exoplanets

The goal of this project is to build and evaluate supervised machine learning models based on their ability to classify KOIs (Kepler Objects of Interest) from the NASA Kepler mission into exoplanets and non-exoplanets using features derived from transit photometry.

"The project aims to automate the exoplanet classification and validation process by learning patterns that distinguish real planetary transits from astrophysical noise and instrumental false positives."

BRIEF DESCRIPTION OF THE DATASET

The dataset is derived from NASA Kepler Mission , specially the Kepler Object of Interest (KOI) catalog called cumulative catalog. Data was acquired by pointing telescope on a target star and it's brightness was captured and any dip in its brightness was recorded. If the brightness dimmed repeatedly then something is passing in front of the star. The shape, depth, time duration and periodicity of the dip provides the information about the object in transit. This is called "Transit Method".
Using these transit features and stellar attributes are used to predict whether the object passing in front of the star was an exoplanet or not.

Data originally contains 50 columns and 9564 rows. Further information on the attributes and features is provided later on.

Acknowledgement: The dataset is taken from NASA Exoplanet Archive (DOI 10.26133/NEA4)

ATTRIBUTES SUMMARY

BRIEF SUMMARY OF FEATURES

Transit Features

Transit properties are the measurable characteristics of the brightness dip that occurs when a planet (or another object) passes in front of its star and blocks part of the star’s light.

- koi_period: How frequently the dip happens.
- koi_duration: How long the dip lasts.
- koi_depth: How much the star's brightness decreases.
- koi_ror: The planet radius divided by the stellar radius.
- koi_prad: The radius of the planet. Product of planet to star (ror) ratio and the stellar radius.

BRIEF SUMMARY OF FEATURES

- koi_impact: "Impact parameter" How central the transit is.
- koi_teq: Approximation for the temperature of the planet.
- koi_insol: "Isolation flux" How much solar energy strikes the planet relative to the earth.
- koi_model_snr: Signal to noise ratio

Stellar Features

Parameters related to the target star. Sometimes the type of star determine if the signal is real or a glitch. It includes:

- koi_steff: Stellar effective temperature of the star in Kelvins.
- koi_slogg: Star surface gravity in base 10 log.
- koi_srad: Star radius.

BRIEF SUMMARY OF FEATURES

Target Variable

The target column: koi_disposition: The label whether the KOI is:

- <u>CONFIRMED:</u> Kepler team has validated it as an exoplanet.
- <u>CANDIDATE:</u> Looks like a planet but not yet confirmed.
- <u>FALSE POSITIVE:</u> The signal is caused by something else.

Although the target variable is not binary we will change this in EDA part of the modelling.

[**Note:** The original dataset contained many columns with processed or model-generated values, as the data had already been partially evaluated by a machine learning pipeline. These columns were removed entirely and are not discussed in this project, as including them could cause data leakage or artificially inflate model performance. Further of these are discussed in next section.]

Data Cleaning and Preparation

- **Data Filtration:** Removed “Candidate” class to train on verified ground truth data only.
- **Leakage Prevention:** Dropped ‘koi_score’ (NASA Confidence) and ‘koi_fpflag’ columns to ensure the model learns physics, not administrative labels.
- **Noise Reduction:** Removed all unnecessary columns (Error bars) to simplify the feature space.
- **Preprocessing:**
  - Dropped the rows with missing values.
  - Applied ‘Standard Scalar’ to normalize features.
  - Split: 80% Training - 20% Testing (Stratified)

METHODOLOGY

Approach: Trained Three distinct classifiers to compare performance across different mathematical foundations.

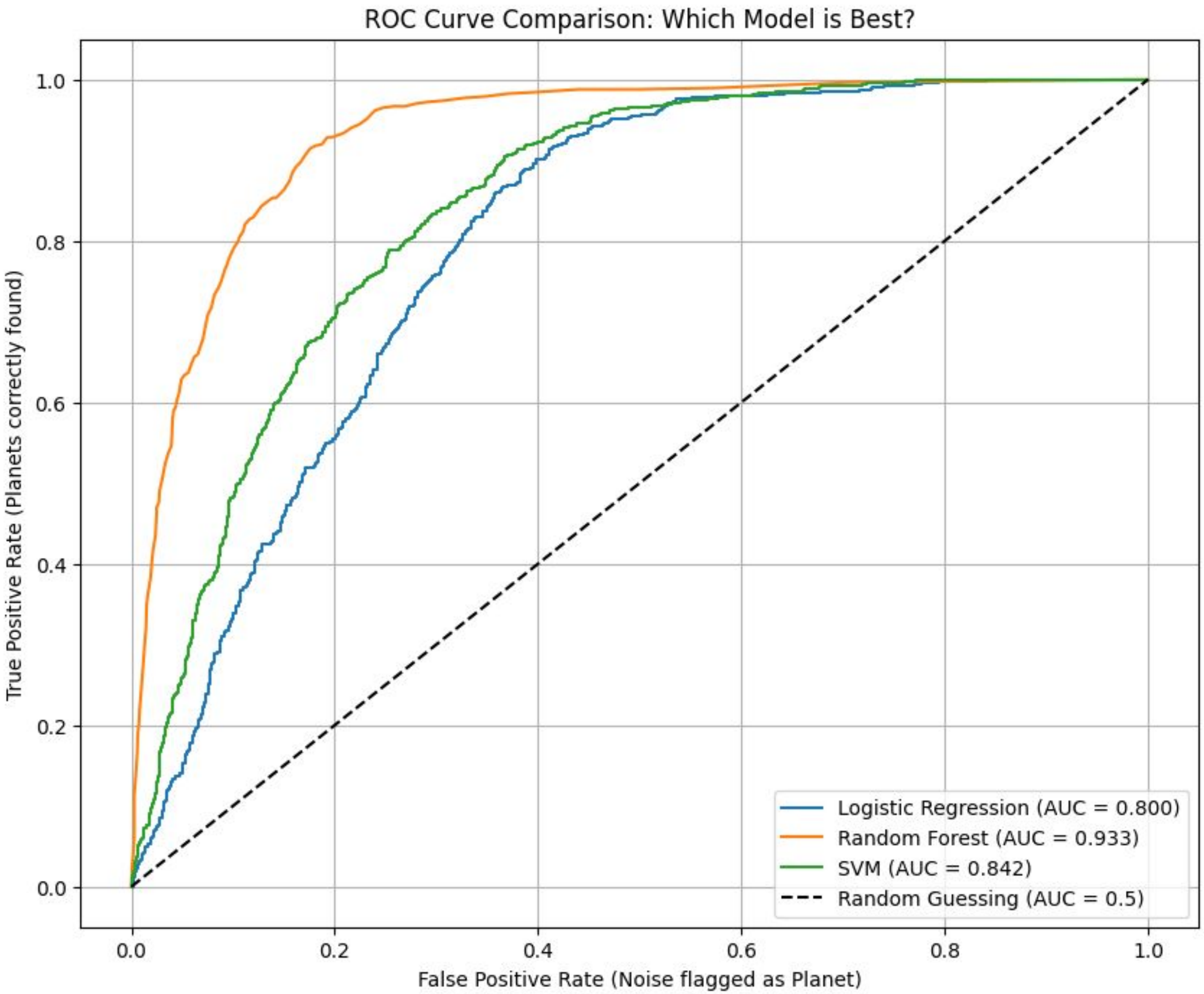
1. **Logistic Regression:** Baseline linear model (highly explainable).
2. **Support Vector Machine (SVM):** Geometric model (good for high dimensional separation).
3. **Random Forest:** Ensemble non-linear model (robust to outliers and complex interactions)

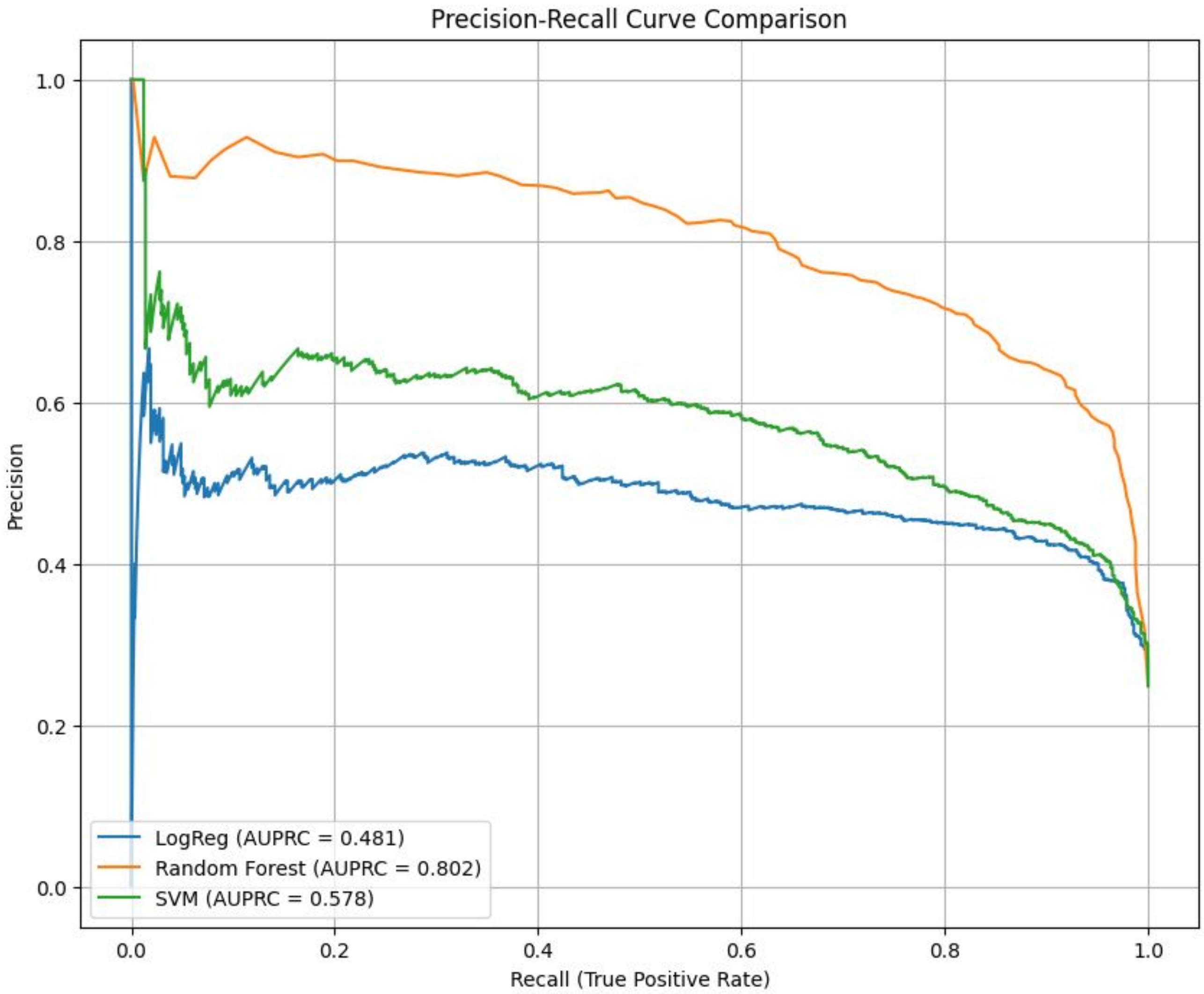
MODEL EVALUATION AND COMPARISON

All three models are trained on the dataset and then evaluated. The classification metrics were used to compare the said models in the table below.

MODEL	ACCURACY	F1 SCORE	AUPRC VALUE
Logistic Regression	75.23%	0.3015	0.481
Support Vector Machine	75.79%	0.1310	0.578
Random Forest	87.03%	0.7322	0.802

As evident, Random Forest performed better than the other two models. It can also be seen in the Precision Recall curve graph and ROC graph on the next slides.


ROC Curve Comparison: Which Model is Best?
True Positive Rate (Planets correctly found)
False Positive Rate (Noise flagged as Planet)
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression (AUC = 0.800)
Random Forest (AUC = 0.933)
SVM (AUC = 0.842)
Random Guessing (AUC = 0.5)


Precision-Recall Curve Comparison
Precision
Recall (True Positive Rate)
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
LogReg (AUPRC = 0.481)
Random Forest (AUPRC = 0.802)
SVM (AUPRC = 0.578)

THE DEEP DIVE

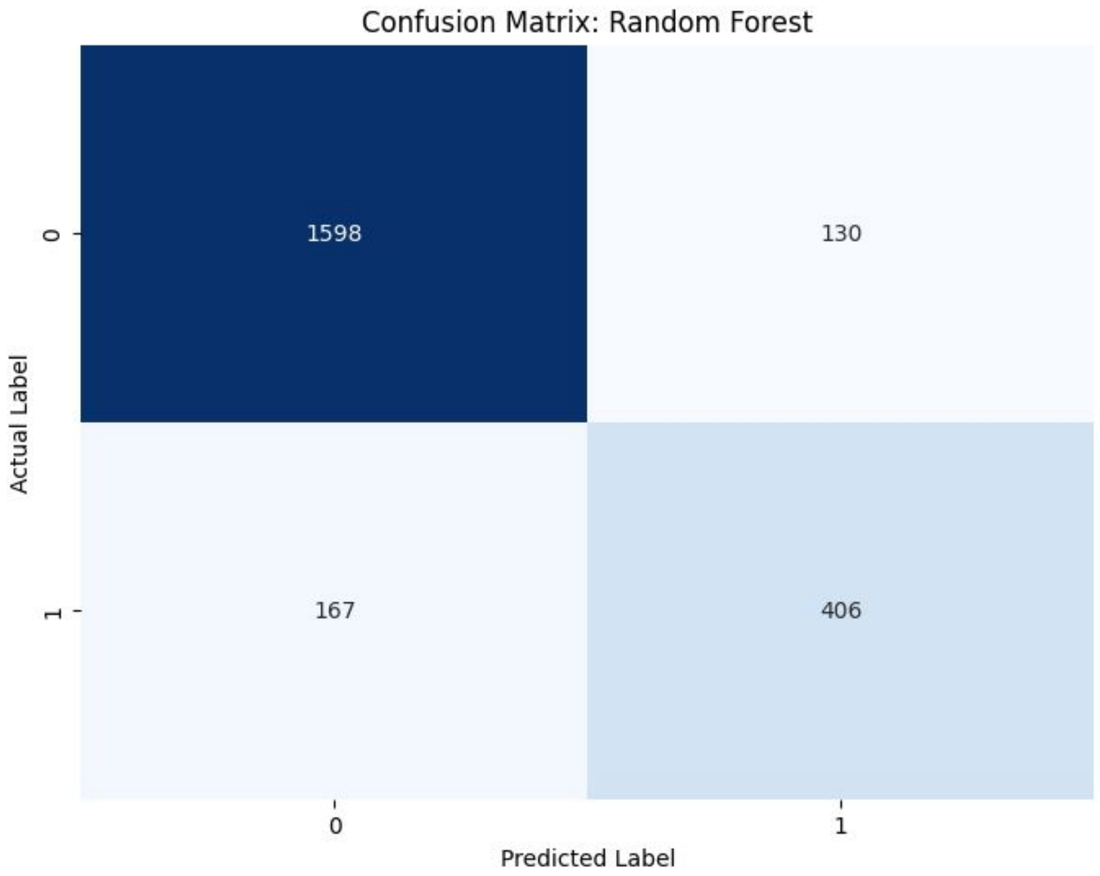
Analyzing the winner model

THE RANDOM FOREST

CONFUSION MATRIX: Random Forest

Confusion Matrix provides a clear summary of model's performance.

- False Positive (top right) number is low, meaning we rarely waste money on fake planets.
- False Negative (bottom left) number is low means precision is high.



KEY INSIGHTS

The Random Forest model with high precision is the best among others. The results are summarized as follows:

- **Precision (85%):** When the model predicts a planet, it is correct 85% of the time, ensuring we rarely waste resources on false leads.
- **Recall (78%):** The model successfully identified 78% of all actual planets in the dataset, capturing the vast majority of discoveries.
- **F1-Score (81%):** With an F1-Score of 81%, the model achieves the optimal balance between filtering out noise and not missing real planets.
- **Accuracy (87%):** Overall, the model correctly classified 87% of all observations, proving it is a highly reliable automated filter.

FEATURE IMPORTANCE

We can deduce which feature was the most important for the model in the process of classification. The feature importance score is used to rank features as follows:

1. **‘koi_model_snr’ (20%):** The signal to noise ratio was the most important feature in deciding whether the KOI was an exoplanet or not.
2. **‘koi_prad’ (~16%):** The planet radius was the second most important feature with 16% importance.
3. **‘koi_impact’ (~10%):** The impact parameter of the transit object is also of grave importance for classification and identification of exoplanets.

RECOMMENDED DEPLOYMENT STRATEGY

The Winner: Random Forest Model Recommended

Justification:

- Superior Performance: Achieved an AUPRC of 0.802, significantly outperforming baseline Logistic Regression (0.481).
- Reliability: High Precision (85%) minimizes false positives, ensuring efficient allocation of the $10M+ annual telescope budget.

Operational Threshold:

- Set the decision threshold at 0.24.
- Why: This maximizes the F1-Score, optimally balancing the cost of wasted resources against the risk of missed discoveries.

FUTURE STEPS & CONCLUSION

Strategic Roadmap

- **Threshold Tuning:** Implement dynamic thresholds based on telescope cost (e.g., stricter for James Webb, looser for ground-based).
- **Data Enrichment:** Incorporate Stellar Metallicity to better predict gas giant formation.
- **Deep Learning:** Explore CNNs to analyze raw light curves directly, rather than just summary statistics.

Conclusion

- **Proof of Concept:** Validated that ML can automate exoplanet detection with **87% accuracy**.
- **Business Impact:** Transforms a manual bottleneck into an automated funnel, accelerating the search for Earth 2.0.

THANK YOU